

Differentiation Digest

$$f'(x) = \frac{d}{dx} f(x) = \frac{df}{dx}$$

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{x \rightarrow a} \frac{f(a) - f(x)}{x - a}$$

Patterns

$$c' = 0$$

$$x' = 1$$

$$(cf(x))' = c \cdot f'(x)$$

$$(x^c)' = cx^{c-1}$$

$$(c^x)' = \ln(c) \cdot c^x$$

$$\log'_c |x| = \frac{1}{\ln(c) \cdot x}$$

$$(f(x) + g(x))' = f'(x) + g'(x)$$

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$(f(x)^{g(x)})' = f(x)^{g(x)} \left(g'(x) \ln(f(x)) + \frac{f'(x)g(x)}{f(x)} \right)$$

$$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx} \Rightarrow \begin{cases} \frac{d}{dx} (f(x))^c = c(f(x))^{c-1} \cdot f'(x) \\ \frac{d}{dx} c^{f(x)} = \ln(c) \cdot c^{f(x)} \cdot f'(x) \\ \frac{d}{dx} \log'_c |f(x)| = \frac{1}{\ln(c) \cdot f(x)} \cdot f'(x) \end{cases}$$

Trigonometric Derivatives

$$\sin'(x) = \cos(x)$$

$$\cos'(x) = -\sin(x)$$

$$\tan'(x) = \sec^2(x)$$

$$\sec'(x) = \sec(x) \tan(x)$$

$$\cot'(x) = -\csc^2(x)$$

$$\csc'(x) = -\csc(x) \cos(x)$$

$$\arcsin'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\arctan'(x) = \frac{1}{1+x^2}$$

$$\operatorname{arcsec}'(x) = \frac{1}{|x| \sqrt{x^2-1}}$$

$$\arccos'(x) = -\arcsin'(x)$$

$$\operatorname{arccot}'(x) = -\arctan'(x)$$

$$\operatorname{arccsc}'(x) = -\operatorname{arcsec}'(x)$$

TODO

- Should trig derivatives be here, or should they just be in the trig digest?
 - hyperbolic trig?
- Does it need e^x and $\ln(x)$ explicitly?
- x^x ?
- Partial derivatives?